



DATA & MACHINE LEARNING CLUB

<DML/>

From Data to Decisions — Building Tomorrow's Innovators Today

📍 Malawi

DATA SCIENCE

Collection • Analysis •
Visualization

MACHINE LEARNING

Models • AI • Real-World
Solutions

INNOVATION

Hardware • Software •
Community

01 EXECUTIVE SUMMARY

The Data & Machine Learning Club (DML Club) is a student-led initiative dedicated to exploring the full cycle of data: from collection and management to analysis, visualization, and the application of machine learning and artificial intelligence.

The club provides a collaborative environment where members gain hands-on experience with hardware, software, and modern data tools, while engaging in projects that address real-world challenges in Malawi and beyond.

The club was established under the mentorship of **Dexter Shepherd, Alejandra Carriero, Paula Lotte Sophia Leonie Seidler** from the **University of Sussex; ISAID Team**. Unlike traditional data science groups, the club teaches members how to work with microcontrollers and sensors that collect data, the applications that manage it, and the ML models that interpret it.

02 VISION & MISSION

🎯 VISION

To become a vibrant community of young innovators using data and technology to drive sustainable development and foster youth empowerment across Malawi and beyond.

🎯 MISSION

To empower students and young professionals with practical skills in data collection, data science, machine learning, and AI, enabling them to design innovative solutions for agriculture, environment, and community development.

03 OBJECTIVES

Our objectives are built around three core pillars:

01

Data Collection

Sensors, microcontrollers,
mobile & web apps.
Digital data platforms.

02

Data Science

Analysis, engineering,
visualization, and insight
from collected and open
datasets.

03

ML & AI

Training and deploying
models using local &
global datasets for
community challenges.

04 ACTIVITIES

The club organizes a range of activities ensuring members gain both theoretical knowledge and practical skills:

- ▶ Workshops and training sessions on sensors, microcontrollers, and coding.
- ▶ Data projects focused on agriculture, environment, and community needs.
- ▶ Applying **ML and AI** to real-world problems.
- ▶ Collaborative projects with institutions, organizations, and communities.

05 EXPECTED OUTCOMES

Through active participation, members will:

- ▶ Develop strong skills in data collection, analysis, and AI applications.
- ▶ Create **practical solutions** for agriculture, environment, and community development.

DATA & MACHINE LEARNING CLUB

- Build a network of students and professionals collaborating on **impactful projects**.

06 LEADERSHIP & MENTORSHIP

ROLE	NAME	AFFILIATION
Facilitator	Martin Chirwa	Co-founder and Hardware Systems
Facilitator	Hobby Bwanali	Co-founder and Software systems
Mentors	Dexter Shepherd, Alejandra Carriero, Paula Lotte Sophia Leonie Seidler	University of Sussex - guiding the club's foundation

07 CONTACT & PARTICIPATION

The Data & ML Club is open to anyone curious about Data, Micro controllers and Machine learning. Members gain hands-on experience moving from “learning by reading” to “learning by building.”

GET IN TOUCH

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